Quantum Memories via Microwave Electron Spin Ensembles

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21 Jul 26

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Why Quantum Memories?

The Scale Problem: Fault-tolerant QC demands unscalable physical footprint

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RSA-2048 via Shor's algo $\sim$ 1M qubits (Gidney, 2025)	Monolithic chip: 13M qubits $\Rightarrow$ $5 \times 5\,\mathrm{m}$ footprint (Alexander, 2023)	**With memory:** Shor's just 13,436 qubits (Gouzien & Sangouard, 2021)

Quantum memory offloads intermediate states from processor, amortises qubit overhead across time

Gouzien & Sangouard's result requires multimode storage times no shorter than $\sim$ 1s

Microwave regime necessary for superconducting compatibility

Spin Ensemble Platform

Key physics: spin ensemble coupled to microwave cavity via collective coupling $g_{\mathrm{eff}} = g_0\sqrt{N}$

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Material platforms:

System	Key property	Role
Si:Bi	Rich hyperfine manifold, microwave transitions	Primary memory
NV/diamond	First coupling of ensemble and SC qubit	Proof of concept
Yb:YSO	$C \geq 1$; microwave transitions at zero field	High-C platform
Er^{3+}:$CaWO_4$	Telecom-band optical transitions	Network bridge

Coherence Engineering: Clock Transitions

Challenge: inhomogeneous broadening spreads spin resonances; *ensemble dephases* faster than spins decohere

Clock transitions: frequencies insensitive to magnetic field fluctuations (*1st order*), dominant noise channel suppressed

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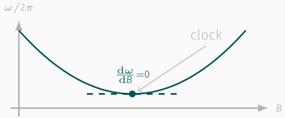
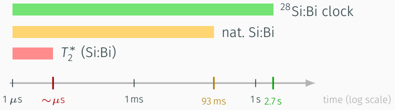


Clock transition: $d\omega/dB=0$ cancels field-noise dephasing

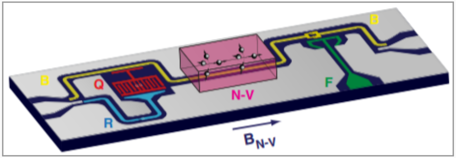


Intrinsic coherence times T_2, T_2^* from *Wolfowicz et al. (2013)*

Hybrid Architecture and Proof of Concept

Two milestones bridge SC-spin coupling to operational memory:

- SC qubit state coherently mapped into spin ensemble and retrieved (Kubo *et al.*, 2011): hybrid architecture validated



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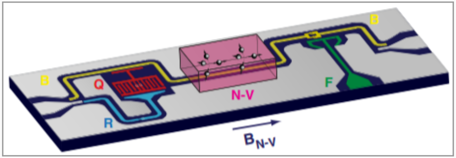

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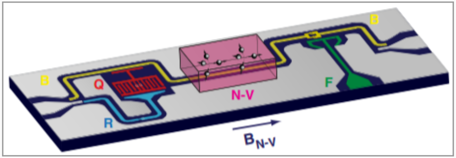

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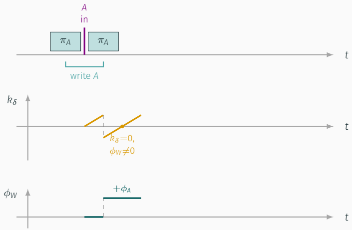
- Hahn echo commits to fixed retrieval order (FILO) and population inversion causes ASE noise

Random-Access Protocol: WURST Pulses

Chirped π-pulse pairs imprint unique phase labels ϕ_W; each label selects a stored mode for readout - giving random access without population inversion



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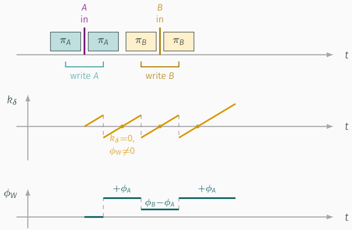

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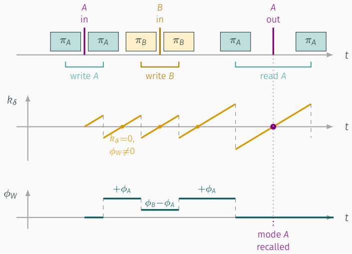

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- *Write B*: (π_B, π_B) stores *B* without disturbing mode *A*
- *Read A*: read-1 π_A cancels ϕ_A; *A* emitted at $k_\delta\!=\!0$; *B* remains stored

WURST: Results

O'Sullivan *et al.* (2022) demonstrated in Si:Bi ensemble:

Metric	Value
Independent modes retrieved on demand	4
Modes resolved within inhomogeneous linewidth	$\sim$16 (theoretical capacity as high as 100)
WURST memory lifetime	2 ms (3$\times$ longer than Hahn echo)

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- Repeated WURST pairs act as **built-in dynamical decoupling**, extending lifetime
- **Random access** achieved: read A while B remains stored
- No population inversion noise (unlike Hahn echo)

Challenges: The Cooperativity Bottleneck

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Routes to unit cooperativity in Si:Bi:

- **Planar microresonator** $\rightarrow$ raise g_0 — O'Sullivan (2020)
- **Isotopic enrichment** (^{28}Si) $\rightarrow$ reduce γ — Wolfowicz (2013)
- **Hyperpolarisation** $\rightarrow$ raise N_{eff} — Morley (2010)

Outlook I: Crossing $C = 1$ with Yb:YSO

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- Simpler level structure + ZEFOZ: no hyperpolarisation needed; narrow γ at zero applied field
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Implication: $C = 1$ removes the fundamental obstacle to high storage efficiency - attention now shifts to losses in the surrounding circuit

Outlook I-B: From Cooperativity to Efficiency

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Path forward: planar microresonator engineering + optimal modulation protocols; unit cooperativity $\rightarrow$ unit efficiency now appears tractable

Outlook II: Erbium Bridge to Optical Networks

Long-term vision: connect microwave spin memories to fibre-optical quantum networks

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Er ions coupled simultaneously to photonic *and* microwave resonator - microwave-to-optical conversion at 1535 nm telecom C-band (Rochman *et al.*, 2023)

Conclusion

Progress in 15 years:

Progress →

NV ensemble coupling (Kubo 2011)	$T_2 = 2.7\,\mathrm{s}$ Si:Bi clock (Wolfowicz 2013)	100 ms storage Si:Bi Hahn (Ranjan 2020)	WURST random access (O'Sullivan 2022)	$C = 1$ Yb:YSO memory (Hogan 2026)

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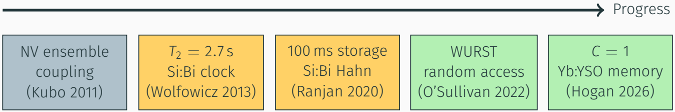

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- **Long-term:** integration of microwave spin memories into fibre-optical quantum networks

Thank you for listening!